GGG-D-280C
May 27, 1988
SUPERSEDING
GGG-D-280B
January 14, 1966

FEDERAL SPECIFICATION

DIE, METAL STAMPING, HAND; AND DIE SET, HAND (LETTER AND NUMERAL)

This Specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers tool steel dies for stamping letters or numerals on castings, forgings, machined components, or weldments for informative and identification purposes. The impressions are made by striking the unmarked end of the die with a hammer.

1.2 Classification.

1.2.1 Types. Dies shall be of the following types as specified (see 6.1):

I - Numerals, heavy duty, metal stamping.
II - Letters, heavy duty, metal stamping.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20407.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; Los Angeles; San Francisco; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, standards, Commercial Item Descriptions from established distributions points in their Agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards

ASTM E18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material used in the manufacture of metal stamping dies shall be of 1095 grade or higher grade alloy tool steel, free from cracks, chips, spalls or other defects and imperfections that may adversely affect the serviceability of the finished product. All dies shall be made from quarter octagon or square bar stock of uniform dimensions.

3.1.1 Hardness. Dies shall show at the character end a Rockwell C hardness of 60 to 63; at the striking end a hardness of 49 to 53; and on the shank midway between the two ends a hardness of 55 to 60 when tested as specified in 4.3.1.

3.2 Design. The characters for metal stamping dies shall be square cut, vertical (Gothic) style, shaped and so located as to be in alignment with the sides of the shank or body and shall be capable of producing neat and legible characters on various materials. The striking end shall be convex and the top edges beveled or chamfered. The character end shall be tapered in accordance with accepted industry practice.

3.3. Marking.

3.3.1 Dies. Each die shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, and country of origin. In addition, the character and the size of the character engraved on the end shall be permanently marked on the "thumb" side of the die, the thumb side being the side adjacent to the base of the character on the end.

3.3.2 Box or cloth roll for sets. Box or cloth roll shall be etched or stamped in a permanent and legible manner with the manufacturer's name or identifying symbol of such known character that the source may be readily identifiable, part number and country of origin. In addition, the box or roll shall be marked in a legible and durable manner with the words "heavy duty".

3.4 Finish and coating. The dies shall be cleaned and all dirt, scale, sharp edges, and burrs shall be removed. A protective coating such as black oxide, black phosphate, electroplating, or lacquer shall be applied to the dies in accordance with accepted industry practice.

3.5 Dies, singles and sets. All dies in a set shall be of the same dimensions in accordance with accepted industry practice. Sets shall be furnished in cloth rolls, plastic boxes (sizes 1/32 inch through 1/4 inch) or wooden boxes as specified in the contract or order (see 6.1). Each box or roll shall contain a complete set of figures or letters, as applicable. There shall be a separate compartment or space provided for each character. The spaces shall allow for clearance so that any piece may be withdrawn or inserted. The boxes shall be of strong construction, and fitted with removable, neat-fitting covers.

3.5.1 Type I, numerals. Type I numerals shall be furnished singly or in sets, as specified (see 6.1). Sets shall consist of numerals 1, 2, 3, 4, 5, 6, 7, 8, and 0 (the numeral 6 being reversed when required to form 9), a total of nine pieces.

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3.5.2 Type II, letters. Type II letters shall be furnished singly or in sets, as specified (see 6.1). All letters shall be capitals. All sets shall contain the complete alphabet and the symbol "&". In addition, sets 1/32-, 1/16-, 1/8-, 3/16-, and 1/4-inch sizes shall contain a "period", making a total of 28 pieces for sets in these five sizes.

3.5.3. Types I and II. Dies shall possess high shock-resisting properties, and shall pass the test specified in 4.3.2. The characters shall be either hand engraved, machine engraved, or machine processed, hand finished and so shaped and supported as to provide maximum strength and durability. Dies shall conform to table I except that a plus or minus tolerance of 0.004 inch will be allowed on the size or height of characters, and commercial tolerances shall apply to the sizes of square bar stock.

TABLE I. Heavy-duty steel dies (numerals and letters)

Minimal size, height of characters Inch	Depth of Characters[1]		Body dimensions			
			Size of square bar stock		Length overall	
	Minimum Inch	Maximum Inch	Minimum Inches	Maximum Inches	Minimum Inches	Maximum Inches
1/32	0.010	0.015	1/4	5/16	2-1/4	3
1/16	.015	.020	1/4	5/16	2-1/4	3
1/8	.025	.030	5/16	3/8	2-1/4	3-1/4
3/16	.035	.040	3/8	7/16	2-1/2	3-1/2
1/4	.040	.045	7/16	1/2	2-3/4	3-1/2
5/16	.040	.045	1/2	9/16	2-3/4	3-1/2
3/8	.045	.050	9/16	5/8	3	3-3/4
7/16	.050	.055	5/8	3/4	3-1/4	4
1/2	.055	.060	3/4	7/8	3-1/4	4
5/8	.055	.060	7/8	1	3-1/2	4-1/4
3/4	.075	.095	1	1-1/8	3-3/4	4-1/2
7/8	.075	.095	1-1/8	1-1/4	3-3/4	4-1/2
1	.100	.125	1-3/8	1-1/2	3-3/4	4-1/2

[1]/Measured from face to base of character.

3.6 Workmanship. Dies shall be free from scale, burrs, and all imperfections which may adversely affect their durability and serviceability. Boxes shall be free from cracks, splinters, or sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.1.2 Inspection lot. All dies of the same type offered for delivery at one time shall be considered a lot for the purpose of lot acceptance inspection and tests.

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4.2 Examination of end item. The sample unit for this examination shall be one die of each type. The inspection level shall be Level I with an acceptable quality level (AQL) of 4.0 percent defective in accordance with MIL-STD-105.

4.2.1 Examination of end item for visual defects and workmanship. The dies shall be examined for nonconformance in design, material, finish, coating, workmanship, and marking.

4.2.2 Examination of end item for dimensional defects. Examination shall be made of the end item to determine compliance with the dimensional requirements specified herein.

4.3 Testing of end item. A random sample of dies shall be selected from each inspection lot for testing hardness and shock-resistance as specified in 4.3.1 and 4.3.2 to determine compliance with the requirements specified therein. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 percent defective in accordance with MIL-STD-105.

4.3.1 Hardness test. Each of the sample dies shall be tested in accordance with ASTM E18 to determine compliance with the hardness requirements of 3.1.1. When the Rockwell reading is within 2 percent of the minimum or maximum allowed, the inspector may at his own discretion, allow the shock-resistance test to be the final determining factor in acceptance or rejection of the lot.

4.3.2 Shock-resisting test. Each of the sample dies shall be subjected to a shock-resisting test. Where practicable, the letter E or F and the figure 2 or 5 should be selected from the 1/8-inch nominal size, unless the inspection lot includes other sizes in considerably larger quantities. The shock-resisting test shall be performed by making repeated legible impressions in material such as heat-treated spring steel or the equivalent, the material to be at least 1/4 inch thick and to have a hardness of not less than 38, nor more than 43 on the Rockwell C scale under a 150-kilogram major load. A 24-ounce machinist's hammer (or any mechanical device, which in the opinion of the inspector will give comparable and satisfactory results), shall be used in making these impressions. The heavy-duty dies under test shall produce at least 2500 impressions under the foregoing conditions without indications of failure. Failure shall be understood to have occurred should the sample chip, spall, or mushroom, the body become bent, or the cutting edges become dulled to an appreciable extent.

4.4 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5 and contract. The sample unit shall be one shipping container fully prepared for delivery. Defects shall be scored in accordance with Table II. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE II. Classification of Preparation For Delivery Defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with PPP-P-40. The level of preservation, packaging and packing shall be A or C, as specified (see 6.2).

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Type, quantity, and whether required singly or in sets (see 1.2, table I, 3.5.1, and 3.5.2).
(c) When required singly, designate the specific letter(s), numeral(s), or symbol(s).
(d) Whether wooden boxes, plastic boxes, or cloth rolls are required for each set (see 3.5).
(e) Level of preservation, packaging, and packing required (see 5.1).

6.2 Type III of GGG-D-280B has been deleted.

MILITARY INTERESTS:

Military Coordinating Activity:
Army - GL

Custodians:
Army - GL
Air Force - 99

Review Activities:
Air Force - 84

PREPARING ACTIVITY:
GSA/FSS

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